

Cambridge International AS & A Level

PHYSICS**9702/32**

Paper 3 Advanced Practical Skills 2

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MARK SCHEME

Maximum Mark: 40

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This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **19** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: - the specific content defined in the mark scheme - the specific skills defined in the mark scheme or in the level descriptions - the standard of response required as exemplified by the standardisation scripts.
2	Examiners must award whole marks (not half marks, or other fractions).
3	Examiners must award marks positively . This means that: - marks are not deducted for errors or omissions - marks are awarded for correct/valid answers as defined in the mark scheme and also for other valid answers which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate - the quality of spelling, punctuation and grammar are judged only when the mark scheme indicates that these features are specifically assessed in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must consistently apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the full range of marks defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and must not award marks with grade thresholds or grade descriptions in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	arithmetic error
AWK	awkward scale used on graph
	benefit of the doubt given
	contradiction in response, mark not awarded
	correct point or mark awarded
	error carried forward applied
FO	false origin used on graph
	incorrect point or mark not awarded
	information missing or insufficient for credit
	power of ten error

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Annotation	Meaning
RO	read-off from graph
SH	supervisor's help given
SR	supervisor's report taken into account
SV	supervisor's value/sample results taken into account
TE	transcription error
IR	value in range
OOR	value out of range

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
SF	Significant figures

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Question	Answer	Marks	Guidance
1(a)	Final value of L in range 3.0–16.0 cm with unit.	1	Ignore precision. If candidate's value is out of range and SV is also out of range, allow $SV \pm 6.0$ cm.
	Final value of L to nearest 0.1 cm. Any values of raw L to nearest 0.1 cm.	1	Allow answer line value for L to the nearest tenth of a mm (0.01 cm) only if repeats of raw values are seen to the nearest mm.
1(b)	Six sets of readings (different values of m) without help from Supervisor scores 4 marks, five sets scores 3 marks, four sets scores 2 marks, three sets scores 1 mark, two or fewer sets scores 0 marks	4	Must be different values of m . Record the total number of sets of readings in a ring to left of table. Allow more than six sets. If help given: 3 marks for six sets of readings, 2 marks for five sets of readings, 1 mark for four sets of readings, 0 mark for three or fewer sets of readings. Indicate with SR on script front, with details near table.
	Values recorded in table for m , L and h .	1	
	Trend correct (L increases as m increases).	1	Judge trend from table <u>not</u> from graph. Not equal values. If one or more values in the table is 'out of trend' then this mark not awarded BUT this may not affect quality mark on the graph yet.
	Range of m : $m_{\min} \leq 150$ g and $m_{\max} \geq 450$ g.	1	Circle minimum and maximum values of m in the table.
	Column headings: Each column heading must contain a quantity and a unit where appropriate. The presentation of quantity and unit must conform to accepted scientific convention e.g. T / cm .	1	Ignore units in the body of the table. Ignore POT errors. Accept separating mark as a solidus, brackets around units or 'in', but not a comma. Do not allow upper case 'G' for unit of m .
	Consistency: All values of raw h must be given to the nearest mm.	1	Do not allow all zeros in the mm position.

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Question	Answer	Marks	Guidance
1(b)	Values of T calculated correctly.	1	Check value in last row of table. Answer should be correct to the number of s.f. given by the candidate. If incorrect from mean L or h values, go back to raw values and recalculate and if still incorrect write in correct value.
1(c)(i)	Axes: Axes must be labelled with the correct quantities. Scales must be chosen so that the plotted points occupy at least half the graph grid in both the x and y directions. Scale markings are no more than 2 cm (one large square) apart. Sensible scales must be used, no awkward scales (e.g. 3:10 or fractions).	1	Accept labels on axes ONLY as T and m. Required labels are T against m. Allow inverted axes m against T. Actual numbers need not correspond to label. Ignore units. Do not allow reversed numerical scales e.g., 10, 9, 8, 7... left to right. Minimum 6 large squares on long axis and 4 large squares on short axis. Ignore point(s) plotted off the grid and assess on remaining points. Allow 8:10. Indicate x-axis false origin with FO. Do not allow non-linear scales (ignore false zeros at origin) Do not allow awkward markings (e.g. 2.57, 3.57, 4.57 an awkward offset of a sensible linear scale)

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Question	Answer	Marks	Guidance
1(c)(i)	Plotting of points: All observations in the table must be plotted on the grid. Diameter of plotted points must be $\leq$ half a small square. Points must be plotted to an accuracy of half a small square in both x and y directions.	1	Ignoring any point off the grid, write a ringed total of points. Indicate reason if mark not awarded. Do not award if: - any table value is not plotted; - numbers do not correspond to axis labels - any point has diameter larger than half a small square (blob); - a point is (or should be) off the grid (check point on edge and check back to table calculation); - the point furthest from the line is plotted more than half a small square out. Ring and check point furthest from the line. Tick if correct, re-plot if incorrect. If point on edge is checked and found correct, still check point furthest from line.
	Quality: General trend of points must be positive. All points in the table must be plotted (at least 5) for this mark to be awarded. It must be possible to draw a straight line that is within ± 0.5 cm on the <i>T</i> axis (normally the y-axis) of all plotted points.	1	Ignore line drawn by the candidate. If examiner has replotted a point judge quality on position of replot. Indicate the tolerance on the graph. <u>Indicate reason</u> if mark not awarded. Do not award if: - wrong trend on the graph (i.e. negative gradient); - a point is (or should be) off the grid (check pt on edge); - non-linear scale in region of the points; - wrong quantity plotted or calculated; 4 or fewer plotted points.

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Question	Answer	Marks	Guidance
1(c)(ii)	Line of best fit: 'Best fit' is judged by the balance of all points on the grid (at least 5 points) about the candidate's line. There must be an even distribution of points either side of the line along the full length. Lines must not be kinked or thicker than half a square. Some candidates may choose to identify an anomalous point. If they identify one point as anomalous (e.g. by circling or labelling) then this point is to be disregarded when judging the line of best fit. There must be at least 5 points left after the anomalous point is disregarded.	1	Allow a straight line through a curved trend. <u>Indicate reason</u> if not awarded. Do not award if: • rotation or translation achieves a better fit; • line is kinked; • a curve has been drawn; • line is thicker than 1 mm or 'hairy'; • points are too scattered to judge; • there are 4 or fewer points. • line does not span all plotted points considered in assessing the line (line must span in either the x or the y direction). Assess line on all points if more than one point is identified as anomalous or only 5 points are plotted on the grid.

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Question	Answer	Marks	Guidance
1(c)(iii)	Gradient: gradient sign on answer line consistent with graph drawn. The hypotenuse of the triangle used should be greater than half the length of the drawn line. Both read-offs must be accurate to half a small square in both the x and y directions. Method of calculation must be correct, not $\Delta x / \Delta y$.	1	Allow hypotenuse to be exactly half the length of the drawn line. <u>Correct substitution must be shown</u>. Do not check beyond the substitution regardless of the method used. If a read-off is incorrect, write in correct value(s). Ignore units. Ignore POT error if it is the same for both values. Allow use of points from the table <u>ONLY</u> if they are less than or equal to half a small square from the line. Check the plotting of these points. Other valid methods: - Using simultaneous equations (two read-offs correctly substituted and sufficiently far apart); - Counting squares $\times$ scale factor (<u>must</u> see a triangle); - Using only Δy and Δx (without read-offs) – check these values to one small square using their triangle (triangle or coordinates <u>must</u> be shown).
	y-intercept: Intercept read directly from the graph, with read-off at $m = 0$, accurate to half a small square in y direction. or Correct read-off from a point on the line is substituted into $y = mx + c$ or an equivalent expression, with read-off accurate to half a small square in both x and y directions.	1	Check for x-axis false origin (FO) i.e. $m \neq 0$. Allow correct intercept read-off from a curved line (for the intercept read directly method only). Allow ecf of gradient value, m. <u>Correct substitution must be shown</u> Check correct substitution in $y = mx + c$. Do not check beyond the substitution. If read-off used is incorrect <u>and gradient mark not awarded because of this incorrect read-off</u>, then check 2nd read-off in the gradient substitution. If this 2nd read-off is also incorrect, do not allow the intercept mark. If correct, allow ecf and award the intercept mark. Ignore POT.

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Question	Answer	Marks	Guidance
1(d)	p equal to candidate's gradient value and q equal to candidate's intercept value. Values must not be written as fractions or to only one significant figure.	1	If $q = 0$, ignore sf in q. Special case: when q is read off the axis: allow 1 sf for q if it is close to the origin and can only be read to 1 sf. Allow POT adjustment if consistent with unit. Ignore any working. If axes are inverted, this mark cannot be awarded unless the candidate makes a correction at this point (true gradient = $1 / \text{candidate's gradient}$ and true intercept = $-\text{candidate's intercept} / \text{candidate's gradient}$).
	Units for p and q correct and consistent with readings (e.g. cm g^{-1} for p and cm for q).	1	Only if $q = 0$, then allow no unit for q. Allow 'cm / g' and 'm / kg' as unit for p. Do not allow upper case 'G^{-1}' or '$/ \text{G}$' in unit for p.

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Question	Answer	Marks	Guidance
2(a)	Final value of H greater than 40.0 cm and to nearest mm. Any values of raw H to nearest mm.	1	Allow answer line value for H to the nearest tenth of a mm (0.01 cm) only if repeats of raw values are seen to the nearest mm.
2(b)	Five values of H , with H increasing as n increases.	1	Tick near table 2.1 on page 8 as this is a data collection mark. Ignore precision for all five values.
2(c)(i)	Final value of $H >$ value of H in 2(a) .	1	
	Evidence of repeat readings of H .	1	
2(c)(ii)	Absolute uncertainty of 2–6 mm and correct method of calculation to obtain percentage uncertainty in H . If repeated readings have been taken, then the absolute uncertainty can be half the range if the working is clearly shown, but NOT zero if values are equal.	1	With $\times 100$ seen or implied. If repeated readings used, must see repeats either here or in 2(c)(i) . Allow new mean H if repeated here rather than in 2(c)(i) .
2(c)(iii)	Value of n to nearest 0.1 paperclips and within 0.1 paperclips of graph line and F correctly calculated from n .	1	Check value of F to number of sf the candidate has used.
2(d)	Final value of d in range 1.0–2.0 cm with unit and to nearest mm. Any raw values of d must be recorded to nearest mm.	1	If candidate's value is out of range and SV is also out of range, allow $SV \pm 0.50$ cm. Allow answer line value for d to the nearest tenth of a mm (0.01 cm) only if repeats of raw values are seen to the nearest mm.
2(e)	Second values for H and d .	1	Any values, ignore units, ignore precision. This value of d must be different from value of d in 2(c)(i) .
	Second H (from 2(e)) $>$ first H (from 2(c)(i)).	1	

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Question	Answer	Marks	Guidance
2(f)(i)	Two values of k calculated correctly. The final k values must not be written as fractions or to only one significant figure.	1	Use values of F and d on page 11 to calculate the second value of k . Check this k value, when rounded to 2 sf is the same as the examiner's value when rounded to 2 sf. Allow k value to 1 sf only if n and F are both shown to 1 sf. Check calculation from raw values if necessary. If incorrect, record correct value. Ignore POTs if same for both k values. Allow your calculated value to be written on either line. If no working shown allow mark if k value correct.
2(f)(ii)	Justification for significant figures in k linked to significant figures in H and d (and W).	1	Allow n and d , as alternative to H and d . Reference to W is not required. Ignore F , ignore 2π . Ignore the actual sf used in the calculation. Do not credit just 'raw readings' if unspecified. Any additional numerical justification must be valid.

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Question	Answer	Marks	Guidance
2(g)	Calculation of percentage difference between candidate's two k values using k values from 2(f)(i). Comparison of percentage difference with 10% leading to a consistent conclusion.	1	1. Look for a % difference calculation. Allow the use of half the % difference calculation e.g. $(k_1 - k_2) / (k_1 + k_2)$. Allow ECF of k values. 2. Comparison to 10(%) or 'the (suggested / percentage) uncertainty'. Do not allow any other uncertainty e.g. their own or the uncertainty in 2(c)(ii). 3. Valid comment linking points 1 and 2. The candidate's conclusion must be consistent with the criterion 10%. Allow use of 10% criterion with one or both k values to determine whether there is an overlap or not, with a valid conclusion stated. Special case: Allow the statement $k_1 = k_2$ only if the values are the same to 3 sf, with a valid conclusion stated.

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Question	Answer	Marks	Guidance
2(h)(i)	A Two (sets of) readings are not enough to draw a valid conclusion. or two k values are not enough to draw a conclusion.	4	Do not allow 'Two readings not enough for (accurate) results'. Do not allow '... to conclude the results'. Allow 'not enough k values to draw a conclusion'. Allow '(one or) two readings are not enough to draw a conclusion'. Allow 'too few readings for a conclusion'. Allow '.... to conclude (the experiment)'. Allow reference to 'relationship' to mean 'conclusion'.
	B Difficult to measure H with reason e.g. parallax, card is moved when touched by rule, card shape is distorted.		Allow 'height' for H . Ignore references to wind. Ignore 'rule not vertical'. Ignore 'rule moves'. Ignore 'difficult to keep rule still'.
	C Difficult to measure d with reason e.g. high % uncertainty, loop not circular, thickness of wire of loop not taken into account.		Allow diameter' for d . Accept 'relative uncertainty' or 'fractional uncertainty' for % uncertainty. Allow 'internal value of d is different to the external value of d '. Ignore parallax errors. Ignore ruler not precise enough. Ignore 'uncertainty' on its own. Ignore 'absolute uncertainty'.
	D Difficulty measuring H at instant when loop just detaches from water surface, with reason e.g. difficult to watch / observe loop and H reading at the same time, difficult to lower the beaker at the same time as reading H , H value changes suddenly.		Allow 'height' for H . Allow 'difficult to watch loop and rule at same time while measuring H '. Allow ' H changes / alters quickly'. Do not allow 'card moves fast/quickly'.

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Question	Answer	Marks	Guidance
2(h)(i)	E Difficulty with apparatus: Card is not horizontal at start or paper clips may not be identical or loop of wire is not horizontal / not in one plane or (larger) loop touches the sides of the beaker.		

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Question	Answer	Marks	Guidance
2(h)(ii)	A Take more readings (for different values of H and d) <u>and</u> plot a graph (H against D) or Take more readings <u>and</u> compare k values.	4	Do not allow 'repeat readings' on its own. Do not allow 'take more readings and (calculate) average k '. Do not allow 'repeat readings and draw a line' without reference to a graph. Allow 'repeat readings and plot a graph'.
	B Clamp rule (to measure H) or Use a thicker card or Use a rectangle made of a named rigid material		Do not allow 'scale' for rule. Allow 'ruler' for 'rule'. Ignore close windows. Ignore turn off fans.
	C Use (vernier / digital) calipers / micrometer / screw gauge (to measure d). or Measure external diameter and measure internal diameter, then find the mean. or Form the loop around a (rigid) cylinder.		Allow 'travelling microscope (to measure d)'. Ignore 'Take several measurements of d (on different diameters) and find mean value'
	D Video (card) with rule (in view) or Use a platform with, e.g. a screw mechanism / motor, to incrementally lower the beaker of water		Do not allow 'scale' for rule. Allow 'ruler' for 'rule'. Allow 'record', 'film' for 'video'. Ignore references to 'timer'. Ignore 'camera'. Ignore 'slow motion'.

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Question	Answer	Marks	Guidance
2(h)(ii)	E Workable improvement to apparatus, e.g.: Add small mass, e.g. adhesive putty, to card to make it horizontal. or Check mass of paper clips using a balance. or A reasonable method to flatten loop. or Use a larger diameter / wider beaker.		 Do not allow 'flatten loop' on its own. Do not allow 'larger beaker' on its own.